SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	P. Sahith Sai	**Reg. No.:**	192425029
Section / Batch:	AIML	**Date:**	22-10-2026

Code of Conduct

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Sahith Sai

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10
Finite-State Morphological Parsing	Morphological Parsing, Finite-State Processing	TP-11

Techniques		
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Morphology and Word Formation in Natural Languages

How words are built from morphemes, and why it matters for NLP systems

Presented By : P. Sahith Sai

Agenda

1. Foundations: words and morphemes
2. Inflection vs. derivation
3. Word formation processes
4. Morphological typology of languages
5. Morphology in NLP: techniques and tools
6. Challenges and future directions

What is Morphology?

- Morphology is the study of the internal structure of words and the rules that govern how they are formed.
- A morpheme is the smallest meaningful unit of a language; a word may contain one or many.
- Morphemes are not syllables: "dogs" has two morphemes (dog + -s) but one syllable.
- Morphology links sound (phonology) and sentence structure (syntax).

un + break + able = unbreakable

Types of Morphemes

Free morphemes

Stand alone as words.
Examples: book, run, happy, the

Bound morphemes

Must attach to another morpheme.
Examples: -s, -ing, un-, -ness

Affix type	Position	Example
Prefix	Before the root	un-happy, re-write
Suffix	After the root	teach-er, quick-ly
Infix	Inside the root	Tagalog: s-um-ulat ("wrote")
Circumfix	Around the root	German: ge-lieb-t ("loved")

Inflection vs. Derivation

Property	Inflection	Derivation
Purpose	Marks grammatical features (tense, number, case)	Creates a new word (lexeme)
Category change	Usually none	Often changes part of speech
Productivity	Highly regular, applies broadly	Less regular, more restricted
Position	Typically outer, after derivation	Typically closer to the root
Examples	walk → walked, cat → cats	happy → happiness, teach → teacher

Order inside a word: root + derivation + inflection, e.g. nation-al-ize-d.

Word Formation Processes

Affixation

Add prefixes or suffixes: disagree, kindness

Compounding

Join free roots: notebook, sunflower

Conversion

Change category with no change in form: to email, a run

Reduplication

Repeat part or all of a word: bye-bye, Indonesian "buku-buku" (books)

Blending

Merge parts of words: brunch, smog

Clipping / Acronyms

Shorten or abbreviate: lab, NASA, laser

Morphological Typology

Type	Idea	Example
Isolating	Few or no affixes; one morpheme per word	Mandarin, Vietnamese
Agglutinative	Many affixes, each with one clear meaning	Turkish: ev-ler-imiz-den = "from our houses"
Fusional	Single affix packs several meanings	Spanish: habl-o ("I speak": 1st person, singular, present)
Polysynthetic	Whole sentences built in one word	Inuktitut, Mohawk

Rich morphology means a huge vocabulary of surface forms, so data sparsity grows.

Why Morphology Matters in NLP

Reduces sparsity

Forms like walk, walks, walked share one lemma; counting them together improves statistics.

Handles OOV words

Unseen words can be understood from known pieces: un-friend-ly.

Improves tasks

Search, machine translation, POS tagging, parsing, speech recognition, spell checking.

Supports low-resource languages

Rule or unsupervised segmentation helps where data is scarce.

Core Morphological Techniques in NLP

Technique	How it works	Trade-off
Stemming	Strips suffixes by rules (Porter, Snowball)	Fast; may over- or under-stem
Lemmatization	Maps a word to its dictionary form using a lexicon and POS	More accurate; needs resources
Finite-state transducers	Encode analysis and generation rules (e.g. Xerox, HFST)	Precise, bidirectional; manual effort
Unsupervised segmentation	Learns morphs from raw text (Morfessor)	No labels needed; noisy
Neural analyzers	Seq2seq / transformer models predict lemma and features	High accuracy; needs data

Stemming vs. Lemmatization: Example

Word	Porter stem	Lemma (with POS)
running	run	run (verb)
studies	studi	study (noun/verb)
better	better	good (adjective)
universities	univers	university (noun)
was	wa	be (verb)

Pipeline: tokenize → analyze morphology (lemma + features) → feed downstream models.

Subword Tokenization (BPE)

- Modern models (BERT, GPT, T5) split words into subword units, not whole words.
- Byte-Pair Encoding (BPE) repeatedly merges the most frequent adjacent symbol pairs.
- Related methods: WordPiece, Unigram Language Model, SentencePiece.
- Result: a fixed vocabulary that still covers rare and new words.

Example: **unbelievably** → **un | believ | ably**
Good alignment here; some splits are purely statistical and ignore morpheme boundaries.

Challenges and Future Directions

Challenges

• Ambiguity: "unlockable" = un-(lockable) or (unlock)-able
• Irregular forms: go/went, mouse/mice
• Non-concatenative morphology: Arabic and Hebrew root-and-pattern (k-t-b)
• Rich morphology with little data
• Code-mixing and evolving words

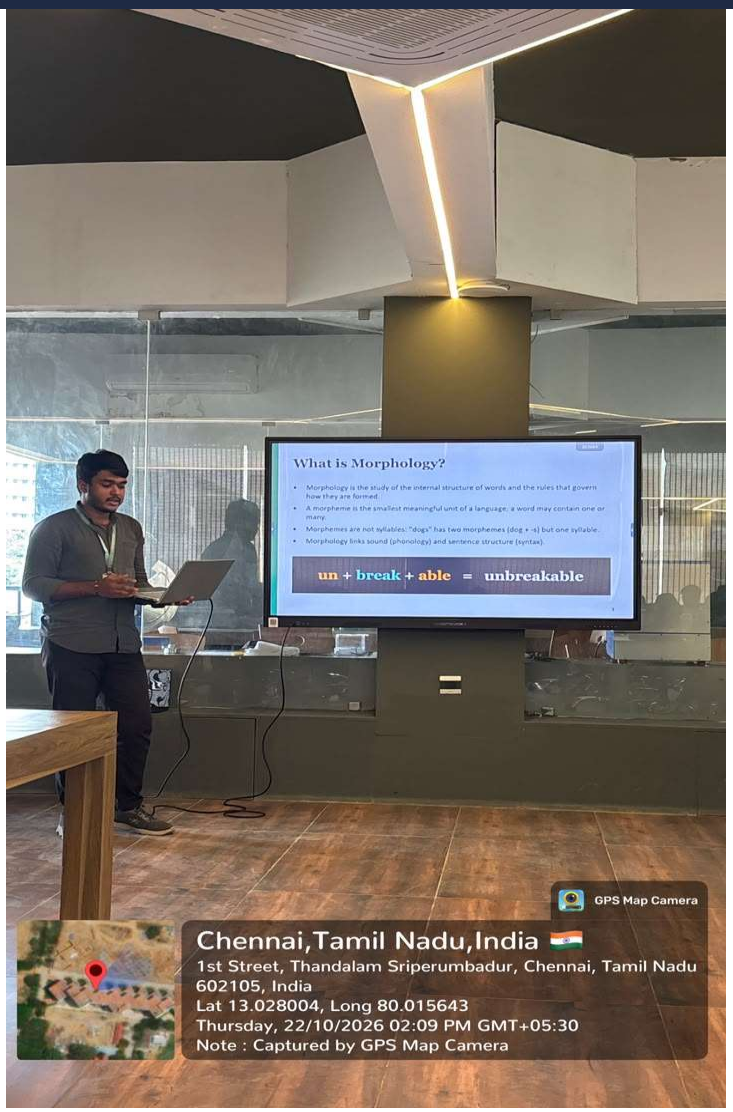
Future directions

• Morphology-aware subword tokenizers
• Multilingual and cross-lingual transfer for morphologically rich languages
• Shared tasks such as SIGMORPHON for inflection and segmentation
• Better evaluation beyond English
• Combining linguistic rules with neural models

Key Takeaways

- Words are built from morphemes via inflection, derivation, compounding and other processes.
- Languages differ widely in morphological type, which shapes NLP difficulty.
- Stemming, lemmatization, FSTs and subword models bring morphology into NLP systems.
- Morphological awareness improves generalization, especially for rich and low-resource languages.

Thank you | Questions?



Rubrics for Calculation

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____